Edge Computing and TinyML for Internet of Things

Enabling Intelligence on Resource-Constrained Devices

From Cloud to Edge: A Comprehensive Survey (2022-2025)

Review Paper

Based on Analysis of Recent Publications

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Abstract

The convergence of edge computing and Tiny Machine Learning (TinyML) is revolutionizing IoT deployments by enabling on-device intelligence without cloud dependency. This paper presents a comprehensive survey of edge AI technologies for resource-constrained IoT devices, covering hardware platforms, software frameworks, and optimization techniques. We examine the evolution from cloud-centric architectures to distributed edge intelligence, analyzing approaches that achieve sub-milliwatt inference on microcontrollers with KB-level memory. Key topics include: (1) edge computing architectures and the cloud-edge-device continuum; (2) TinyML frameworks (TensorFlow Lite Micro, Edge Impulse, microTVM); (3) model compression techniques including quantization, pruning, and knowledge distillation; (4) hardware accelerators for edge AI; (5) distributed ML pipelines across IoT networks. Our analysis reveals that optimized TinyML models can achieve 90%+ accuracy while consuming less than 1mW power, enabling always-on sensing applications. The InTec framework demonstrates effective distribution of ML pipelines between edge and cloud. We identify challenges in model accuracy-efficiency trade-offs, heterogeneous deployment, and over-the-air updates. Future directions point toward neuromorphic computing, in-sensor processing, and automated edge ML optimization.

Keywords: Edge Computing, TinyML, IoT, Microcontrollers, Model Compression, Edge AI, Embedded Machine Learning, Low-Power AI

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1 Introduction

1.1 The Edge Computing Paradigm

Traditional IoT architectures relied heavily on cloud computing for data processing and intelligence. However, this approach faces significant challenges:

- **Latency**: Round-trip cloud communication introduces unacceptable delays for real-time applications
- **Bandwidth**: Transmitting raw sensor data consumes limited network resources
- **Privacy**: Sensitive data exposure during cloud transmission
- **Reliability**: Dependence on network connectivity
- **Cost**: Cloud computing expenses scale with data volume

Edge computing addresses these challenges by processing data closer to its source, while TinyML extends this capability to the most resource-constrained devices.

1.2 TinyML: Machine Learning at the Edge

TinyML represents the deployment of machine learning models on microcontrollers (MCUs) with:

- Memory: 256KB - 2MB RAM
- Processing: 50-400 MHz ARM Cortex-M cores
- Power: Microwatts to milliwatts
- Cost: $1-10 per device

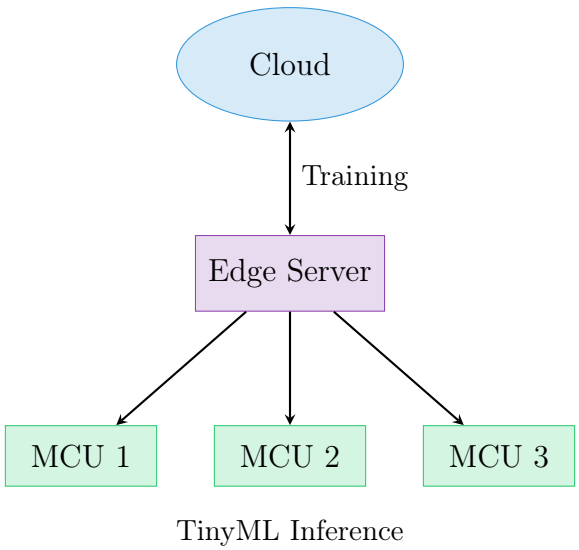


Figure 1: Cloud-Edge-Device Architecture for TinyML Deployment

2 Edge Computing Architectures

2.1 Hierarchical Edge Computing

Modern IoT deployments adopt multi-tier architectures:

Table 1: Edge Computing Hierarchy

Tier	Latency	Compute	Examples
Cloud	100+ ms	High	AWS, Azure, GCP
Edge Server	10-50 ms	Medium-High	NVIDIA Jetson, RPi
Gateway	1-10 ms	Medium	Industrial gateways
Device (TinyML)	<1 ms	Low	STM32, ESP32

2.2 InTec: Integrated Things-Edge Computing

The InTec framework provides a systematic approach to distributing ML pipelines across the edge-cloud continuum:

- Automatic partitioning of ML models between edge and cloud
- Dynamic workload balancing based on resource availability
- Support for heterogeneous hardware platforms
- Optimized data transmission between tiers

2.3 Edge AI Taxonomy

A comprehensive taxonomy of Edge AI includes:

1. **Inference-only edge**: Pre-trained models deployed on edge
2. **Training-capable edge**: On-device fine-tuning and adaptation
3. **Collaborative edge**: Distributed learning across edge nodes
4. **Autonomous edge**: Self-optimizing edge systems

3 TinyML Frameworks and Tools

3.1 Software Frameworks

Major TinyML Frameworks

- **TensorFlow Lite Micro**: Google's framework for MCU deployment
- **Edge Impulse**: End-to-end TinyML development platform
- **microTVM**: Apache TVM for microcontroller compilation
- **CMSIS-NN**: ARM's optimized neural network kernels
- **STM32Cube.AI**: ST's neural network deployment tool

3.2 Model Compression Techniques

To fit ML models on constrained devices:

Table 2: Model Compression Techniques Comparison

Technique	Size Reduction	Accuracy Loss	Method
INT8 Quantization	4x	0.5-2%	Reduce precision
Pruning	2-10x	1-3%	Remove weights
Knowledge Distillation	5-20x	2-5%	Teacher-student
Neural Architecture Search	Variable	Minimal	AutoML

3.3 Hardware Platforms

Table 3: Popular TinyML Hardware Platforms

Platform	Memory	Power	ML Accelerator
Arduino Nano 33 BLE	256KB	5mW	No
ESP32-S3	512KB	20mW	Vector instructions
STM32H7	1MB	50mW	No
MAX78000	512KB	1mW	CNN accelerator
GAP9	1.5MB	10mW	Cluster + NPU

4 Applications and Use Cases

4.1 Indoor Environmental Monitoring

Edge computing methods have been extensively evaluated for indoor parameter estimation, comparing approaches like:

- On-device TinyML inference
- Edge server processing

- Hybrid edge-cloud solutions

Results show that edge-based approaches reduce latency by 10-100x while maintaining prediction accuracy within 2% of cloud solutions.

4.2 Object Detection at the Edge

Benchmarking deep learning models on edge devices reveals:

- MobileNet-SSD achieves 20+ FPS on Jetson Nano
- YOLO-Tiny provides real-time detection on Raspberry Pi 4
- Quantized models reduce memory by 4x with <3% accuracy loss

4.3 Machine Learning In-Sensors

Emerging paradigm of computation-enabled sensors:

- Processing integrated directly in sensor hardware
- Near-zero latency for inference
- Ultra-low power consumption (microwatts)
- Applications in always-on keyword detection, gesture recognition

5 Challenges and Solutions

5.1 Accuracy-Efficiency Trade-offs

Key Trade-offs

- Model size vs. accuracy
- Inference latency vs. energy consumption
- Generalization vs. specialization
- Update frequency vs. stability

5.2 Heterogeneous Deployment

Challenges in deploying across diverse hardware:

- Different instruction sets and capabilities
- Varying memory architectures
- Hardware-specific optimizations required
- Testing and validation complexity

5.3 Over-the-Air Updates

Maintaining edge ML models requires:

- Secure model distribution
- Incremental update mechanisms
- Rollback capabilities
- Bandwidth-efficient transfer protocols

6 Future Directions

1. **Neuromorphic Edge Computing**: Spiking neural networks for event-driven, ultra-low-power inference
2. **In-Memory Computing**: Processing within memory arrays to eliminate data movement
3. **Automated TinyML**: AutoML for edge-optimized architecture search
4. **Federated Edge Learning**: Collaborative training across distributed edge devices
5. **Self-Adaptive Edge AI**: Models that automatically adapt to changing conditions

7 Conclusion

Edge computing and TinyML are transforming IoT by enabling intelligent processing on resource-constrained devices. The progression from cloud-dependent to edge-native architectures has achieved remarkable efficiency gains, with modern TinyML solutions consuming less than 1mW while maintaining high accuracy. Frameworks like InTec demonstrate effective distribution of ML workloads, while hardware advances continue to expand the capability envelope. Key challenges remain in balancing accuracy with efficiency and managing heterogeneous deployments. The future points toward neuromorphic computing, in-sensor processing, and automated optimization as enabling technologies for ubiquitous edge intelligence.

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